Predicting NFL Play-Calling: Modeling Run vs. Pass Tendencies Under Strict Pre-Lineup Constraints

Caden Pascual
capascual@ucsd.edu
University of California, San Diego
La Jolla, CA, USA

Rishil Chintalapati
rchintalapati@ucsd.edu
University of California, San Diego
La Jolla, CA, USA

Figure 1: NFL players set at the line of scrimmage prior to the snap.

Abstract

In the National Football League (NFL), predicting offensive-play calling requires understanding complex, situational game contexts. In this study, we use modern NFL play-by-play data from the 2022 to 2024 seasons to predict whether an offensive team will execute a run or a pass play. By engineering features such as down, distance to go, score differential, and field position, we establish a Logistic Regression baseline model. We then build on this using advanced non-linear models, such as Random Forest, XGBoost and LightGBM. Our best model uses LightGBM to predict plays against unseen data from the 2025 season, achieving a 70.0% accuracy and 73.2% F1-score, proving that game context heavily influences offensive decision-making.

Keywords

Machine Learning, NFL, Sports Analytics, Predictive Modeling, Feature Selection

ACM Reference Format:
Caden Pascual and Rishil Chintalapati. 2026. Predicting NFL Play-Calling: Modeling Run vs. Pass Tendencies Under Strict Pre-Lineup Constraints. In *Proceedings of Data Mining (DSC 148).* ACM, New York, NY, USA, 7 pages.

DSC 148, San Diego, CA
2026.

1 Introduction

Play-calling in football is a complex, strategic battle between the offense and defense. Historically, teams have relied on coaching intuition and film study to predict the opponents next move, the recent "analytics revolution" has changed this dynamic. Football teams now employ dedicated data science teams to optimize these reads, incorporating data-driven algorithms into their decision-making.

Despite the highly dynamic nature of football, historical data shows underlying patterns that can reveal offensive tendencies. If the defense can accurately predict the probability of a run or a pass before the play, they're able to take advantage through personnel substitution and defensive coverage.

This project applies data mining techniques to uncover these tendencies. We seek to accurately classify unseen offensive plays as either a run or a pass based strictly on pre-lineup variables. The remainder of this paper is structured as follows: Section 2 reviews related work in football analytics; Section 3 defines our predictive task and dataset; Section 4 outlines our baseline and advanced modeling methodology; Section 5 presents our final evaluation; Section 6 discusses limitations; and Section 7 concludes our findings.

2 Related Work

The landscape of football analytics has expanded rapidly due to open-source play-by-play data like `nflfastR` [1]. Foundational research by Yurko et al. [3] pioneered the use of this highly-detailed data to establish reproducible frameworks for evaluating game state and situational play value. Building on this foundation, researchers use predictive modeling to capture coaching behavior. Fernandez et al. [2] successfully applied decision-tree ensembles, including Random Forests and Gradient Boosting, demonstrating that non-linear models significantly outperform traditional linear baselines by automatically capturing complex interactions between features like down, distance, and field position.

Our work builds directly on this lineage but explicitly addresses two major gaps prevalent in the existing literature:

(1) **Elimination of Information Leakage:** Many play-prediction models rely on post-lineup features (e.g. *shotgun* alignments or *no-huddle* tempos) to artificially maximize accuracy scores [2]. Because defensive coordinators must substitute personnel and call defensive schemes before the offense sets its formation, we limit our feature space to pre-lineup variables to ensure real-world utility.

(2) **Temporal Evaluation:** Rather than using standard randomized k-fold cross-validation, which allows seasonal/coaching tendencies to leak across data splits, we use a strict temporal split. We train our models exclusively on the 2022–2024 seasons and validate prediction on the 2025 NFL season.

3 Dataset and Exploratory Data Analysis

The dataset consists of NFL play-by-play data sourced via the `nflfastR` library. To prevent data leakage, we utilized a time-based split: the 2022–2024 seasons act as our training data, while the 2025 season is reserved for final testing.

3.1 Data Cleaning

Non-action plays, such as timeouts, penalties, and quarter breaks, were filtered out. The target variable, `play_type`, was mapped to a binary format where pass plays equal 1 and run plays equal 0.

3.2 Feature Selection

We restrict the model to information available before the offense lines up, since our goal is to predict the call as a defensive coordinator would, while the play is still being signaled in. We keep the situational fields that most directly drive real play-calling decisions:

- **Down and Distance:** `down` (one-hot) and `ydstogo`. Third and long almost forces a pass, while short yardage invites a run, so the call depends sharply on this pair.
- **Field Position:** `yardline_100` and `goal_to_go`. Teams backed up near their own end zone tend to play it safe, and the red zone changes the calculus again.
- **Game Context:** score differential, quarter, time remaining in the half and in the game, and timeouts. A team down two scores late throws on nearly every down to save the clock, while a team protecting a lead runs to drain it.

- **Market Signals:** pre-snap win probability and the Vegas spread and total. Heavy favorites expect to lead and lean on the run, while games with high totals tend to be pass-heavy shootouts.

We one-hot encode down because pass rate is non-monotonic across downs (it climbs through third down, then falls on fourth). We also engineer interaction features that make this football logic explicit, including indicators for two-minute drills, short and long yardage, third and long, and the red zone, plus a continuous urgency term, `score_time_pressure`, equal to the score differential divided by the minutes remaining.

We deliberately exclude pre-snap formation cues such as shotgun and no-huddle. A defense can read these once the offense lines up, but a coordinator calling the play earlier cannot, so using them would be look-ahead leakage for our task. They are also largely a consequence of the same situation we already encode, since a trailing team in a two-minute drill lines up in shotgun precisely because it intends to pass. We quantify their value separately as an ablation in Section 4. After cleaning, this yields 105,681 training plays and 34,503 test plays, with offenses passing on roughly 57% of snaps.

3.3 Exploratory Data Analysis

During our exploratory data analysis (EDA), we looked to quantify the foundational football schemes utilized on offense. We segmented key continuous features to evaluate how run-pass distributions shift under specific situational constraints.

First, we analyzed the distance for a first down (`ydstogo`), splitting our data in four distinct bins: Short (1-3 yards), Medium (4-7 yards), Long (8-10 yards), and Very Long (greater than 10 yards). Upon conducting bivariate analysis, a difference in offensive tactics was found between each of these groups. As shown in Figure 2, there is a direct correlation between yardage and probability of pass; however, this trend breaks down at the Long bin. Although this group requires a larger yardage distance than the Medium one, the probability of a running play is higher in the Long bin compared to the Medium bin. This is because of the inclusion of precisely 10 yards in the Long category, which contains many first down plays where teams attempt to run.

Next, we incorporated a `game_script` variable that captures the game state based on the classification of

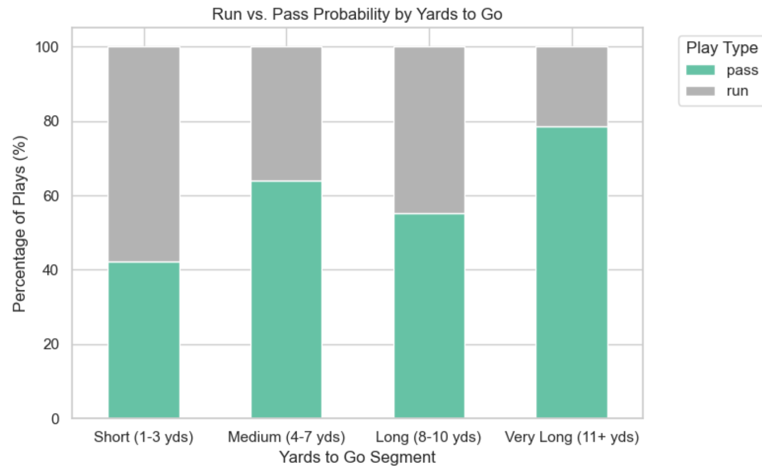

Figure 2: Run vs. Pass Probability segmented by Yards to Go.

`score_differential` into one of three states: "Trailing Big" (more than 8 points behind), "Close Game" (8 or fewer points between teams), and "Leading Big" (more than 8 points ahead). Refer to Table 1 for an example of the effect of `score_differential` on play-calling. Close games see a natural base-passing tendency (56.4%), while games where a team is losing multiple touchdowns have a much higher pass percentage of 68.2%, due to teams' effort to score quickly. In contrast, "Leading Big" games see this pass percentage drop significantly to 46.1%, while runs become more frequent at 53.9%, due to teams wanting to run down the game clock.

Table 1: Offensive Play-Calling Tendencies Based on Game Script

Game Script	Pass (%)	Run (%)
Trailing Big	68.2	31.8
Close Game	56.4	43.6
Leading Big	46.1	53.9

4 Methodology and Modeling

4.1 Baseline Model

We use two baselines. The simplest, a majority-class predictor that always guesses pass, fixes the accuracy floor at the league pass rate of 56.9% and has no discriminative power (AUC 0.50). The second is Logistic

Regression, the canonical linear classifier, which models the log-odds of a pass as a weighted sum of the standardized features. This treats each factor as an independent, additive nudge, which is a reasonable first cut but cannot capture the conditional logic coaches actually use: third and one and third and fifteen point to opposite calls even though they share a down. On the 2025 test set logistic regression reaches 67.9% accuracy, already eleven points above the majority baseline, confirming that play calling is substantially predictable from the situation alone. Any advanced model must clear this bar to justify its added complexity.

4.2 Advanced Modeling

Real play calling hinges on combinations of conditions, for example third and long while trailing late is almost always a pass, but second and short with a lead is usually a run, so we turn to tree-based models that learn these interactions automatically. A single Decision Tree captures them but overfits when grown without limit, so it mainly serves as a building block. We then use two ensembling strategies from the course: a Random Forest, which reduces variance by averaging many bootstrapped, feature-subsampled trees (bagging), and gradient boosting, which reduces bias by fitting trees sequentially to the errors of the current ensemble. We evaluate two boosting libraries, LightGBM and XGBoost, to test whether the choice matters. On the 2025 test set the three ensembles are statistically tied (69.7% to 70.0% accuracy), with LightGBM marginally ahead, and the near-identical LightGBM and XGBoost results match the common finding that the boosting library is immaterial on tabular data once both are tuned. We also evaluated Gaussian Naive Bayes, which trails the field because its assumption that features are independent is plainly false here: down, distance, score, and win probability all move together. We deploy the tuned LightGBM as our final model.

4.3 Parameter Sensitivity

We tuned hyperparameters with 5-fold cross-validated Grid Search and additionally swept individual parameters to visualize the bias-variance trade-off. For Logistic Regression we varied the inverse regularization strength C; for the Random Forest we varied `max_depth` and `n_estimators`; and for gradient boosting we varied

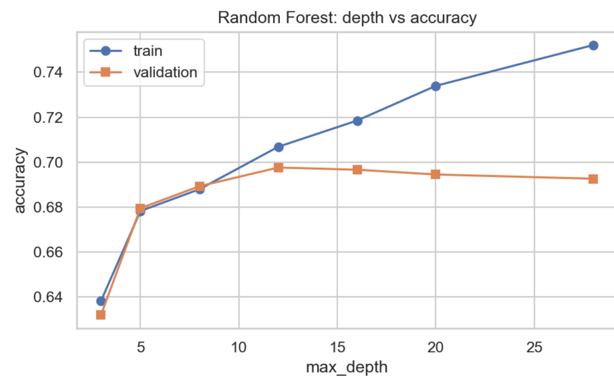

Figure 3: Random Forest sensitivity to tree depth. Training accuracy keeps rising while validation accuracy plateaus and then falls, the signature of overfitting.

the learning rate and the number of leaves. The validation curves show the textbook pattern (Figure 3): as the trees are allowed to grow deeper they start memorizing quirks of individual 2022 to 2023 games rather than general tendencies, so training accuracy keeps climbing while validation accuracy plateaus near a depth of 12 and then declines. The Grid Search selected a learning rate of 0.03, 400 trees, 31 leaves, and a minimum of 50 samples per leaf, achieving 69.7% cross-validated accuracy. Because the models train in seconds on roughly 100,000 rows, scalability was not a constraint.

5 Results and Final Evaluation

The final model was evaluated on the strictly unseen 2025 test dataset.

5.1 Performance Metrics

We report Accuracy, Precision, Recall, and F1 to account for the class imbalance. Table 2 compares all models on the 2025 test set. Every model beats the majority baseline, and the gradient-boosted LightGBM is best, reaching 70.0% accuracy and 0.732 F1, an improvement of 13.1 points over the baseline. The confusion matrix (Figure 5) shows balanced errors across runs and passes rather than a collapse onto the majority class. The remaining error is partly irreducible: offenses deliberately stay unpredictable to counter defensive scheming, so a fraction of calls is intentionally random and cannot be recovered from situation alone.

Table 2: Model performance on the 2025 test set (values are percentages).

Model	Acc.	Prec.	Rec.	F1
Majority (always pass)	56.9	56.9	100.0	72.5
Logistic Regression	67.9	71.2	73.2	72.2
Random Forest	69.7	74.9	70.4	72.6
XGBoost	69.7	73.6	72.8	73.2
LightGBM	**70.0**	**74.5**	**72.0**	**73.2**

We ran two ablations. First, adding the excluded pre-snap formation cues raises test accuracy from 70.0% to 72.5% (Figure 4). This is exactly the edge a defense gains by waiting to see the offense line up, since a shotgun, no-huddle look strongly signals a pass. Second, we tested encoding team identity as each team's historical pass rate. It added 0.5 points on the 2024 validation set but vanished on the 2025 test set (a 0.1-point loss). The reason is practical: a team's tendency is set by people who turn over from year to year, not by the franchise label. A new head coach or coordinator can flip a run-first team into a pass-first one overnight; a quarterback change matters even more, because a team with a weaker or younger passer leans on the run to protect him, while an elite quarterback invites more throwing; and injuries, trades, and free-agency losses (a workhorse running back or a number one receiver leaving) reshuffle the balance. We confirmed this directly: a team's pass rate correlates only 0.37 between consecutive seasons overall, but 0.70 when the coach and starting quarterback both stay and just 0.26 when either changes, with quarterback changes driving the largest swings. Team identity is therefore a stale proxy, which is why we leave it out of the deployed model.

5.2 Feature Importance

Figure 6 shows the gain-based feature importances of the deployed model. Field position (`yardline_100`), the game clock (`half_seconds_remaining`), our engineered `score_time_pressure` term, and yards to go are the most influential, followed by the market signals. This lines up with how coaches actually think: they manage the clock and the scoreboard first, throwing to catch up or running to close out a game, and they weigh field position and distance before deciding how aggressive to be. That the engineered urgency feature ranks among the top three validates our feature design,

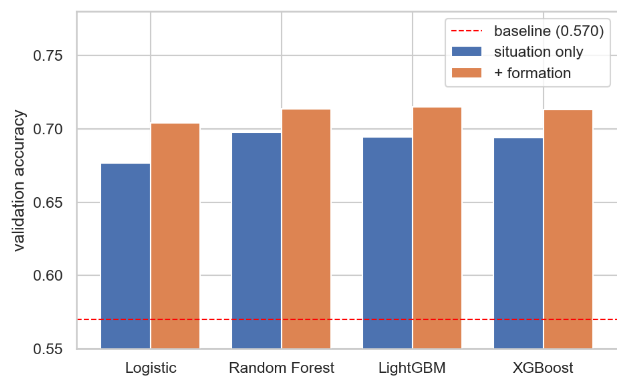

Figure 4: Ablation on the 2024 validation set: adding post-lineup formation cues lifts accuracy consistently across models, the edge a defense gains by waiting for the lineup.

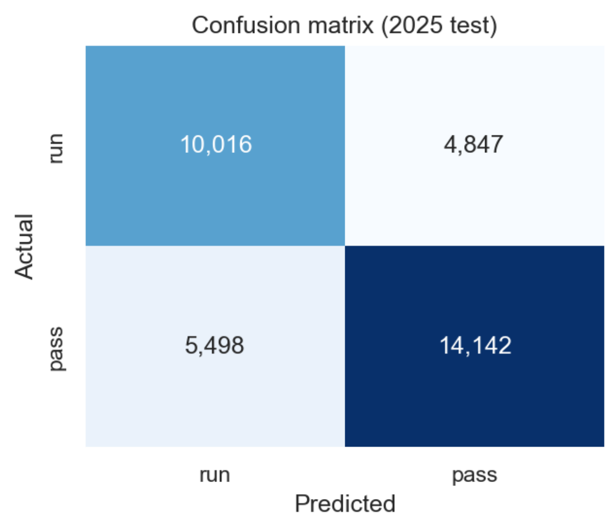

Figure 5: Confusion matrix for the deployed model on the unseen 2025 test season. Errors are balanced across the two classes.

since it packages the comeback and clock-management logic that would otherwise be split across several raw columns. The influence of down is spread across its one-hot indicators rather than concentrated in a single bar, which is why it does not top the chart despite being a strong predictor.

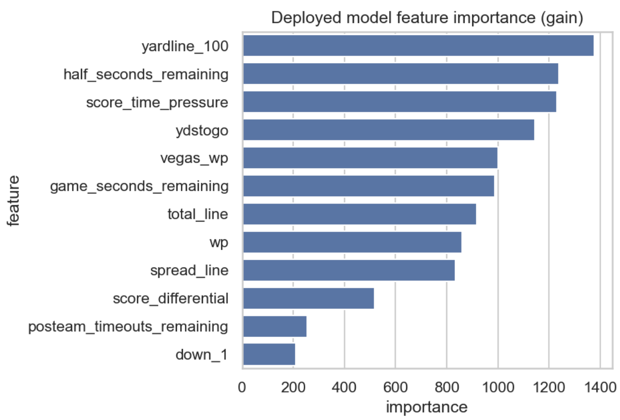

Figure 6: Top features of the deployed model by gain. Field position, game clock, the engineered urgency term, and yards to go dominate.

6 Discussion and Limitations

Several factors bound how high any model can score on this task. Run versus pass is a coarse label, and coordinators reason at the level of specific concepts, so two very different plays can share our target. More fundamentally, good offenses are deliberately unpredictable: they vary their calls to counter what the defense shows, so a portion of play selection is intentionally random and cannot be recovered from the situation alone. This sets a ceiling well below 100 percent and explains why even the best models plateau near 70 percent. Our analysis also has limitations: we infer quarterback changes from the most frequent passer and coaching changes from the recorded sideline coach, so injuries to backups and mid-season changes are captured imperfectly, and we exclude personnel groupings and pre-snap motion because they are not available before the lineup and would reintroduce the leakage we are trying to avoid.

The most promising next step follows directly from why our team feature failed. Because the franchise label is a stale proxy for the players and coaches who actually set tendencies, richer and dynamically updating data should help. Player-level grades and participation data from a provider such as Pro Football Focus (PFF) would let us replace the static team pass rate with current measures of personnel quality: a team's quarterback grade, its offensive line run- and pass-blocking grades, and the quality of its backs and receivers. Crucially, these update as rosters change, so they would capture the very injuries, trades, and quarterback swings that broke the team feature rather than freezing last season's identity. A team that loses its starting quarterback would see its grade-based features shift the same week, where the team label would not. We would similarly encode the current offensive coordinator and his career pass rate, keying the model on the actual play-caller instead of the franchise.

A second direction is to model the matchup. A leakage-safe summary of the opposing defense, such as its historical blitz rate, coverage tendencies, and run-stop grades from PFF, would let the model account for how a specific defense pushes an offense toward the run or the pass. Third, drive context is informative: the previous play, the current tempo, and whether the offense just ran play-action all shift the next call, and a sequence model could exploit this. Finally, the cleanest framing of our whole study is a decision-timeline model that predicts at several moments, before the huddle breaks, after personnel is announced, and once the formation is set, to measure exactly how much information the defense gains at each step. Player-tracking data such as the NFL's Next Gen Stats would make that analysis possible.

7 Conclusion

In this study, we successfully demonstrated that NFL offensive play-calling is highly predictable using strictly pre-lineup situational data. By engineering features that capture football logic, such as score-time pressure and specific yardage segments, our final deployed LightGBM model achieved an accuracy of 70.0% and an F1-score of 73.2% on an unseen 2025 test dataset. This represents a 13.1 percentage point improvement over the baseline league pass rate.

Our feature importance analysis confirmed that coaching decisions are heavily influenced by game clock, field position, and distance to the first down marker. While offenses will intentionally mix up their play-calling to throw off the defense, our results demonstrate that the mathematical game script can classify the vast majority of strategic decisions on the field.

References

[1] Sebastian Carl and Ben Baldwin. 2020. *nflfastR: Functions to Efficiently Access NFL Play by Play Data.* https://github.com/nflverse/nflfastr R package version 4.0.0.

[2] C. Fernandez, R. Yakubov, Y. Li, A. Prasad, and T. C. Y. Chan. 2020. Predicting plays in the National Football League. *Journal of Sports Analytics* 6, 1 (2020), 35–43.

[3] Ronald Yurko, Samuel Ventura, and Maksim Horowitz. 2019. nflWAR: A reproducible method for offensive player evaluation in football. *Journal of Quantitative Analysis in Sports* 15, 3 (2019), 163–183.